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Predict and optimize: a smart inventory management model for beauty retail

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Abstract

Effective inventory management is critical for retail companies, particularly in the beauty sector, where balancing stock availability and cost control directly impacts profitability. While inventory management is recognized as a key driver of operational efficiency, it faces challenges since demand fluctuates and future needs are highly uncertain. In this paper, we examine the inventory management problem for a beauty retailer, and show how we can improve upon their current approach to inventory management, by combining stochastic optimization methods with modern prediction approaches. Currently, the company uses a periodic review system, in which inventory levels are reviewed at set intervals and replenishment orders are based on current stock levels and historical demand distribution. Although this approach is cost-effective, it struggles to adapt to changes in demand over time and periodic patterns within the replenishment cycle and during lead times, which can result in lost sales or overstocking. Our model enhances the traditional periodic review system by integrating stochastic demand forecasting and dynamic re-optimization, allowing for more responsive and informed ordering decisions. By accounting for demand uncertainty and adjusting inventory levels accordingly, the model aims to minimize overall operating costs while maximizing service levels. For a beauty retailer, where customer loyalty is a key consideration, we show that the new method leads to higher service levels, which reduces inventory costs and increases customer satisfaction.

Keywords Inventory management · Two-stage stochastic programming · Uncertain demand · Periodic review inventory policy · LightGBM forecasting · Scenario prediction

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1 Introduction

Inventory management is an essential part of supply chain operations, directly influencing a company's ability to meet customer demand, maintain profitability, and achieve operational efficiency. One of the main challenges companies face is determining the optimal balance between stock held to mitigate against the risk of stockouts, and the need to reduce excess inventory in order to reduce costs and optimize overall business operations [1]. Two fundamental aspects of inventory management are the determination of when to place an order and how much to order. The Economic Order Quantity (EOQ), introduced by Harris, 1913 [2] assumes constant demand and no uncertainty in supply. However, real-world inventory systems face demand variability, compounded by market trends and changing consumer preferences. This complexity makes inventory management a critical focus for businesses wanting to stay competitive and responsive in a dynamic market.

To manage the stochastic and uncertainty in demand, Bookbinder and Tan [3] introduced a framework for inventory control strategies, comprising three key approaches: static uncertainty (also known as the (Q, r) policy), dynamic uncertainty (the (s, S) policy), and static-dynamic uncertainty (the (r, S) policy). They categorize these strategies based on when replenishment decisions are made, either at the start of the planning horizon or after observing demand in a given period.

For example: The (Q, r) policy is one where the inventory level is reviewed continuously, which typically means daily, and a replenishment order of size Q is made when the total inventory on hand, and any stock on order, is less than the reorder point, r. Q and r are typically determined to minimize total reorder, holding and lost sales or back-order costs, see, for example, [4–7].

The (s, S) policy, referred to as the dynamic uncertainty strategy by [3], is the most cost optimal approach, as proven by Scarf [8]. In this strategy, ordering decisions, are made after assessing actual demand. If the inventory position drops to or below s on the replenishment cycle, the order is made to meet the target S. This makes the policy highly effective in managing unpredictable demand changes. However, as demonstrated by Tunc et al. [9] and De Kok and Inderfurth [10], the policy is subject to setup-oriented nervousness, where frequent adjustments to order timing can reduce its practicality.

The (r, S) policy, which Bookbinder and Tan classify as a static-dynamic uncertainty strategy, sets a fixed replenishment periods at the beginning of the planning horizon but allows the order quantities to be determined when the actual orders are placed. In the (r, S) policy, replenishment is predetermined and fixed at the beginning of the period. This approach is well-suited for industries that prefer fixed ordering schedules and need to establish sales, production or shipment plans in advance as is the situation for the case study company. This approach can be more popular as it minimizes setup-oriented nervousness, as highlighted by [9], and allows for better arrangements with carriers due to a predictable ordering schedule. One of its limitations is the difficulty in adapting to real-world situations, particularly when lead times increase.

Although the inventory models described above are designed to minimize total inventory costs in the presence of uncertain demand, they have some significant limitations. These models rely on demand distribution estimates to guide decisions, which may not always accurately reflect real-world variability. Although the periodic review model is popular in real-world companies, because it is reactive to demand uncertainty, it lacks the flexibility in integrating detailed future demand forecasts. This limitation restricts its ability to adapt to strategic factors such as marketing-driven demand fluctuations or seasonal trends.

The case study company is a multi-brand retailer, and needs to order products from multiple overseas suppliers. It faces several challenges that significantly affect inventory management decisions. First, orders must be placed following a fixed replenishment cycle, restricting the retailer's ability to respond dynamically to fluctuations in demand.[1] Second, due to the long lead times associated with international shipments, the retailer faces increased risks of lost sales or overstocking, particularly in the presence of demand uncertainty. These constraints necessitate an optimized inventory replenishment strategy that accounts for variability in demand, lead time, and operational constraints while ensuring customer demand fulfillment and maintaining high service levels. For the case study company, maintaining high service levels for their customers is of significant importance, often outweighing cost considerations. Consistently meeting customer demand without excessive delays or stockouts promotes trust and long-term customer relationships, which are critical for maintaining competitiveness in the retail industry. Therefore, the ability to predict demand fluctuations and make informed replenishment decisions is essential to prevent lost sales and customer dissatisfaction.

The challenge is further complicated by the inability of these approaches to account for dynamic factors such as seasonal variations in demand, peak demand at Easter and Christmas, or shifts driven by marketing trends. Moreover, while stochastic optimization excels at managing demand uncertainties over the planning horizon, it is less effective in adapting to short-term demand fluctuations. As a result, inventory managers are increasingly seeking models that integrate more accurate and dynamic demand forecasts, enabling them to reduce overall inventory costs and improve operational efficiency [11].

To address these challenges of inventory planning for the case study company, this paper proposes a novel approach that integrates machine learning-based forecasting with stochastic optimization in a periodic review inventory system. The methodology combines the LightGBM [12] model to predict future demand, using quantile regression to generate daily demand scenarios over the review interval for the stochastic optimization model. Although inventory levels are updated daily, ordering decisions are made only at fixed review intervals, aligning with the periodic review strategy currently used by the case study company. The proposed approach combines predictive accuracy and operational efficiency, allowing for timely forecast updates and informed inventory decisions, while maintaining the periodic review framework.

[1] Emergency orders are possible but very expensive.

The structure of the paper is as follows: Sects.2 and 3.1 introduce the problem and the key elements of our proposed method: the inventory replenishment cycle and replenishment decisions, and implementation of our stochastic optimization model. The design of experiments section, Sect. 4, describes the customer demand data used as inputs into our experiments, inventory operating protocols, performance measures and the design of our trials. Sections 5.1 and 6 illustrates the results and sensitivity analysis. Finally, conclusions and managerial insights are provided in Sect. 8.

2 Problem description

The case study company is a beauty retail company operating in Australia. They currently use a monthly review system for inventory management, where the ordering quantities are determined at regular intervals based on historical data. However, the primary challenge arises from the international supply chain, where most of the beauty products are sourced from overseas suppliers. This creates a significant lead time of up to 30 days from the suppliers to the warehouse, which adds considerable complexity to the ordering process. Given these long lead times and the uncertainty of demand, it is crucial for the company to determine the optimal ordering quantities to avoid both stock-outs and overstocking.

A key issue in this case study is the difficulty in accurately predicting future demand. Demand in the beauty retail sector can vary significantly, influenced by factors such as changing consumer preferences, seasonal trends, and promotional activities. With a fixed review cycle and long lead times, the company faces a critical challenge in adjusting their inventory levels dynamically to meet demand variations. Ordering too little can result in stock-outs, leading to lost sales and decreased customer satisfaction, while ordering too much can lead to excess inventory, increased holding costs, and the risk of obsolescence.

The ability to determine the right order quantities at the right time is fundamental to optimizing the entire supply chain and enhancing operational efficiency. If the company can accurately predict future demand and determine the optimal reorder point and order quantity, it will be better positioned to strike the right balance between customer satisfaction and cost control. This involves minimizing the risks of both stockouts and overstocking, thereby maximizing service levels, reducing unnecessary costs, and improving the overall profitability of the business. The goal of this study is to develop a model that not only predicts future demand but also implements an optimal inventory plan to effectively manage demand uncertainty, providing the company with a more flexible, responsive, and efficient inventory management strategy, taking into account limited data availability, unpredictable demand, and operational constraints such as long supply lead times.

2.1 Inventory replenishment policy

The case study company employs a **periodic review, order-up-to-level and review interval**, (r, S), inventory policy, whereby replenishment orders are made at fixed

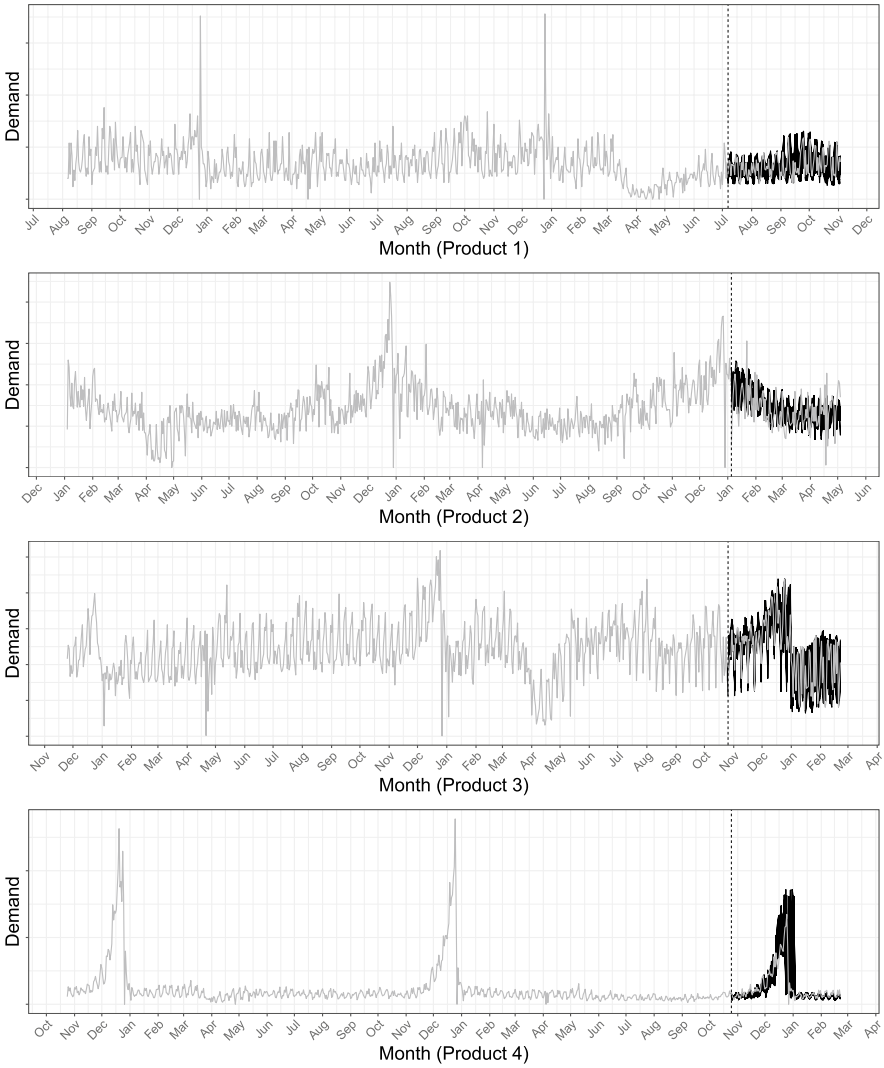


Fig. 1 LightGBM forecasts used in models for Products 1, 2, 3, and 4. The figure illustrates historical demand (730 days) and "future" demand (120 days) in gray, alongside future scenarios (black) generated through random sampling from forecast quantiles for each time step. Note that the displayed time periods correspond to varying date ranges, causing Christmas peaks to appear at different positions

intervals, r, with the goal of achieving a target inventory level, S, when the goods ordered are received after a fixed lead time, l, [7].

This policy is particularly effective when demand is stochastic and stationary, to minimize total holding costs, stockout risks, and ordering costs over time when replenishment orders required to be made at specific time intervals (for example, monthly) [5, 13]. Figure 1 shows that the products studied exhibit fluctuating

demand over time but a relatively constant mean over each planning/replenishment period, making them suitable for this modeling approach. The figure shows there is an annual peak in each demand time series corresponding to Christmas sales, and we adjust for this using a dynamic variant of the inventory policy in the experiments that follow.

The relevant inventory parameters in the model that follows are: holding cost, H, being the cost of carrying stock held in inventory and calculated as the cost per item held in inventory per day at the end of each day's normal inventory cycle; lost sales cost, B, is the penalty of failing to supply any item within the period it is required, and is calculated as the number of items required but not supplied during each inventory cycle; Reordering cost, A, is the fixed expense incurred each time an order is placed, reflecting the logistics and handling costs associated with replenishment.

Demand on any given day, t, represented as D_t, follows a stochastic distribution having mean μ_D and standard deviation σ_D. Inventory levels fluctuate over time, with I_t denoting inventory on hand at time t and so_t representing the number of stock-outs on day t. r is the number of days in the review period, that is, the time interval between replenishment orders being made.

In the experiments that follow, it is assumed that lead time and review period are the same, thus $l = r$, which is similar to the experience of the case study company, although l and r may differ in practice. The (r, S) policy is implemented in this case by calculating the order-up-to-level, S, using a heuristic, attributed to Wagner, [14], and given in [5], as follows:

1. The Economic Order Quantity is calculated as $Q_E = \sqrt{\frac{2\mu_D A}{H}}$.
2. A Safety Factor, u, is determined as $G(u) = \frac{HQ_E}{B\sqrt{r+1}\sigma_D}$, where $G(u)$ is the standard normal loss function.
3. If $Q_E > 1.5\mu$, then $S = (r+1)\mu_D + u\sqrt{r+1}\sigma_D + Q_E$, otherwise
4. The alternative safety factor v is calculated as $\Phi(v) = \frac{B}{B+H}$, where Φ represents the standard normal cumulative distribution, and $S = (r+1)\mu_D + \min\left[u\sqrt{r+1}\sigma_D + Q_E, v\sqrt{r+1}\sigma_D\right]$

Forecast-driven Periodic Review The heuristic above is based on long term historical estimates of μ_D and σ_D. We term this a *static* implementation of the periodic review model. To better reflect seasonal peaks and trends over the longer term, we also implement a *forecast-driven* periodic review model in the experiments that follow by recalculating μ_D and σ_D at each review period using predicted future demand over a planning horizon of twice the review period, r.

3 Stochastic optimization approach

The inventory ordering challenge is a type of two-stage stochastic optimization problem. It involves creating an ordering strategy in the face of uncertain future demands, which become known only after the ordering decisions have been

made. Because the replenishment decisions for one cycle influence those of subsequent days, the problem we address involves determining a sequence of orders for a defined planning horizon, spanning from day t_1 to day T.

The deterministic equivalent formulation (refer to [15] for more details) is a widely used technique for addressing two-stage stochastic optimization problems, particularly in cases involving integer decision variables.

This approach represents uncertain demand through a collection of scenarios, denoted by $j \in J$, where each scenario has an associated probability P_j, satisfying $\sum_{j \in J} P_j = 1$. Each scenario provides a forecasted daily demand, $D_{t,j}$, for each day t within the inventory planning horizon, T. The ordering decisions are made at intervals determined by the review period, r.

In the first stage, the main decision variables are Q_t, which represents the quantity ordered on replenishment day t given a lead time of l days. An auxiliary variable, rec_t, is used to define the quantity received each day in terms of Q. Another binary variable, y_t, indicates whether a non-zero order is placed on replenishment day t.

For the second stage, the decision variables include the inventory level ($I_{t,j}$) and stock-outs ($so_{t,j}$) at the end of each day t for a given scenario j. The ordering decisions O_t for the l days before the optimization period, $t_1 - l \leq t < t_1$, are treated as fixed and based on historical data. The current stock level is denoted by IS.

The model used to determine the ordering decisions for the planning period $t_1, \ldots, T$ is:

$$\min \sum_{t_1 \leq t \leq T} (Ay_t) \quad + \sum_{j \in J} P_j \sum_{t_1 \leq t \leq T} (H * I_{t,j} + Bso_{t,j}) \tag{1}$$

subject to:

$$rec_t = O_{t-l} \qquad \forall\ t_1 \leq t < t_1 + l \tag{2}$$

$$rec_t = Q_{t-l} \qquad \forall\ t_1 + l \leq t \leq T \tag{3}$$

$$(I_{t,j} - so_{t,j}) = I_{t-1,j} + rec_t - D_{t,j} \qquad \forall t_1 < t \leq T, j \in J \tag{4}$$

$$(I_{t_1,j} - so_{t_1,j}) = IS + rec_{t_1} - D_{t_1,j} \qquad \forall j \in J \tag{5}$$

$$y_t = 0 \qquad \forall t_1 \leq t \leq T\ , t \mod r \neq 0 \tag{6}$$

$$\begin{aligned} &Q_t \leq My_t && \forall t_1 \leq t \leq T \\ &I_{t,j}, so_{t,j}, Q_t, rec_t \in \mathcal{Z}^{\geq 0} && t_1 \leq t \leq T, j \in J \\ &y_t \in \{0,1\} && \forall t_1 \leq t \leq T \end{aligned} \tag{7}$$

The objective is to minimize the reorder cost combined with the weighted sum of holding costs and stock-out costs across all scenarios, as defined in line (1).

Equation (2) specifies that the amount received during the initial l time steps is determined by the historical order quantities. Similarly, Eq. 3) sets the amount received later in the planning period to be equal to the order quantity placed l days earlier.

Equation (4) updates the inventory level at the end of day t for each scenario j by adding the received products and subtracting the demand. Since both $I_{t,j}$ and $so_{t,j}$ are constrained to be non-negative, this equation also calculates $so_{t,j}$ as the quantity of stock-outs, if any, that occur. Equation (5) computes inventory levels for the first day, t_1. The same principle applies, but it uses the initial stock level IS instead.

Equation (6) enforces the ordering decision to align with the fixed replenishment cycle based on the review interval.

Finally, Eq. 7) ensures that y_t is set to 1 whenever a positive reorder quantity is placed.

The stochastic optimization function for inventory decisions during the period t_1 to T, given the scenarios $\bar{P}$, is defined as $\mathsf{IM}(r, l, H, A, \bar{D}, IS, \bar{O}, s, \bar{P}, t_1, T)$. This function returns the vector of order quantities $\bar{Q} = [Q_{t_1}, \ldots, Q_T]$, which constitutes the optimal solution to the model described above.

A limitation of the stochastic optimization model is that it lacks any motivation to maintain inventory at the conclusion of the planning horizon, T. An optimal solution will try to drive this to zero to lessen holding costs.

3.1 Predict and optimise

This section explains how prediction techniques are integrated into a rolling-horizon stochastic optimization framework to develop a highly responsive method for managing periodic review inventory.

Traditional deterministic equivalent models rely on demand distributions to generate potential future scenarios and make static inventory planning decisions over a fixed horizon. However, this paper introduces a dynamic methodology that involves making periodic order decisions by incorporating real-time demand and updating inventory levels on each day. This approach improves the accuracy of decisions for future periods.

The stochastic optimization model operates over a rolling horizon, making decisions for a time interval from t_1 to $t_1 + W$, where W is defined as the *planning window*. Ordering decisions are made according to a fixed replenishment schedule on a periodic review policy.

At the end of each day, actual demand is observed, and inventory levels are updated accordingly. The planning window then shifts forward, from $t_2 = t_1 + 1$ to $t_2 + W$. Replenishment orders are made every r days to fulfill anticipated future demand, and are expected to arrive after a lead time of l days. This process is repeated daily, updating inventory levels and advancing the planning window until all decisions for the entire period $1 \ldots T$ have been finalized.

A key advantage of this rolling horizon approach is its adaptability to real-time demand changes. By continuously updating inventory levels, the model overcomes the limitations of static methods, which rely on limited information such as mean

and standard deviation. This dynamic process allows the model to make more accurate predictions for the next 60 days, ensuring that inventory levels are neither excessive nor insufficient, thus minimizing lost sales. As a result, the model becomes more sensitive to market changes, providing a clearer and more informed outlook for the case study company.

A limitation of the traditional stochastic optimization model is its reliance on fixed demand distributions to generate future scenarios. However, with the emergence of modern, highly efficient, and accurate demand prediction methods, such as LightGBM ([12]), we can adopt a more flexible and adaptive approach to scenario generation.

LightGBM is a highly effective gradient boosting framework that has earned widespread recognition for its exceptional performance in machine learning applications, including the renowned *M5* Forecasting competition especially for inventory management sector [16]. It is frequently used for tasks such as time series forecasting, parameter estimation, and integration with ARIMA-type models [17]. It has proven valuable in scenarios with limited knowledge of data patterns [18] or when combined with other machine learning techniques, such as Long Short-Term Memory (LSTM) neural networks [19].

Traditional time series forecasting methods, such as ARIMA and ARMA, face significant limitations that restrict their effectiveness in addressing complex, real-world challenges. These models rely on the assumption of data stationarity, often requiring transformations that can distort the underlying relationships within the data. They also struggle with capturing non-linear patterns, handling high-dimensional features, and managing datasets with missing or sparse values. Additionally, these methods are computationally expensive for large datasets due to their reliance on iterative parameter estimation processes.

By contrast, LightGBM excels at efficiently processing large datasets, high-dimensional inputs, and complex feature structures, making it particularly suitable for forecasting applications involving sparse or categorical features frequently observed in real-world problems. In this study, we selected LightGBM for its ability to quickly generate precise and accurate forecasts during each iteration of stochastic optimization. This capability offers a significant advantage over traditional methods by enabling rapid adaptation to dynamic changes in the data. [20]

Furthermore, the success of LightGBM in the M5 competition highlights its effectiveness in demand forecasting, where speed and accuracy are crucial for adaptive inventory management. Unlike ARIMA and ARMA, LightGBM excels at identifying nonlinear relationships and interactions between features, allowing it to deliver superior predictive performance in diverse and challenging forecasting environments. This makes it an ideal choice for our application, ensuring robust and efficient forecasting outcomes.

A LightGBM model is trained on historical demand data using features such as “day of series" (representing the sequential day number in the time series), “day of year," “day of week," and “month of year." The data is used as-is, without any preprocessing steps such as differencing or smoothing.

This implementation is carried out in Python using the Lightgbm library ([20]), with default parameters applied. Quantile regression forecasts are

then generated at deciles $0.1, 0.2, \ldots, 0.9$ from the fitted model. For each forecast period, scenarios are created by randomly sampling a single value uniformly from the quantile set at each period. Historical data and the corresponding forecast scenarios for each product are illustrated in Fig. 1, demonstrating this method's ability to effectively capture trends and variability.

For each rolling horizon, the LightGBM quantile regressor generates j demand scenarios to use in the stochastic optimization model. Its capability to quickly predict future demand based on historical data and quantile regression makes it an efficient tool for scenario generation.

Algorithm 1 Pseudocode for Periodic Replenishment Predict and optimize model

Input: $r, H, A, \bar{D}, IS, \bar{O}, j, t_1, T, W$
Output: $\bar{Q}$
$t \leftarrow t_1$
$\bar{Q} \leftarrow \bar{O}$
while $t \leq T$ **do**
  if $t \bmod r = 0$ **then**
    Generate a LightGBM model $\mathbf{L}$ using demand data $D_{t-730}, \ldots, D_{t-1}$
    Generate j scenarios $\bar{P}$ for days $t, \ldots, t+W-1$ using $\mathbf{L}$
    $\bar{Z} \leftarrow \mathsf{IM}(r, r, H, A, \bar{D}, IS, \bar{Q}, \bar{P}, j, t, t+W-1)$
    $Q_t \leftarrow Z_t$
  else
    $Q_t \leftarrow 0$
  end if
  $IS \leftarrow \max(0, IS - D_t + Q_{t-l})$
  $t \leftarrow t+1$
end while
return $\bar{Q}$

The pseudocode in Algorithm 1 describes a periodic replenishment approach for stochastic optimization, combining daily demand forecasting with periodic inventory management. The algorithm iterates over time steps $t_1 \leq t \leq T$, where replenishment order decisions are updated only on specific fixed days determined by $t \bmod r = 0$, where r is the predefined replenishment interval. At each time step, a LightGBM model ($\mathbf{L}$) is trained using demand data from the past two years to generate s demand scenarios ($\bar{P}$) for the subsequent W-day planning window. On replenishment days ($t \bmod r = 0$), a stochastic optimization model computes the optimal replenishment quantities ($\bar{Z}$), and the first decision (Z_t) is implemented as the order for the day (Q_t). On non-replenishment days, no order is placed ($Q_t = 0$). The inventory stock level (IS) is updated daily by considering actual demand and past orders. This approach balances computational efficiency through periodic ordering while maintaining adaptability via daily demand updates and scenario-based optimization. Once the horizon T is reached, the accumulated order history ($\bar{Q}$) is returned.

4 Design of experiments

Customer demand data was provided by the case study company for several anonymized products having differing demand profiles. These products present varying trends over time, seasonal demand patterns (such as peaks during weekends and pre-Christmas shopping periods), and days of no sales, often aligning with public holidays. This variety is used to assess the model's performance in diverse real-world applications. Figure 1 provides an example of typical demand patterns for the products across multiple years, highlighting weekly and annual cycles, as well as long-term trends. Below, we present a brief overview of the products analyzed.

- *Product 1* has variations in its weekly trend, while minimal annual seasonality. Peak demand is nearly three times higher than the average. The average daily demand is approximately 700 units, with a standard deviation of about 70 units per day. The dataset covers a period of roughly 10 years, including approximately 100 days with no recorded sales.
- *Product 2* features a stable trend with noticeable weekly seasonality. Annual seasonality is particularly strong, characterized by significantly lower demand during the winter months. Peak demand is about five times higher than the average. The average daily demand is ≈ 50 units, with a standard deviation of ≈ 30 units per day. The dataset spans around 8 years and includes approximately 20 days with no recorded sales.
- *Product 3* presents an increasing trend with very little weekly or annual seasonality. Overall, demand is highly variable with peak demand being approximately twice the average. The average daily demand is ≈ 125 units, with a standard deviation of approximately ≈ 100 units per day. The dataset covers a span of nearly 10 years and includes around 20 days without recorded sales.
- *Product 4* has a stable trend with significant annual seasonality and moderate weekly patterns. Peak demand reaches nearly ten times the average. The average daily demand is ≈ 100 units, with a standard deviation of about 120 units per day. The dataset spans roughly 5 years, with around 10 days where no sales were recorded.

Operating costs The case study company provided us with estimates of the following costs, which reflect real-world operational expenses. All costs are in Australian Dollars, AUD. The *reorder cost (A)*, which includes the expenses associated with placing, receiving, and stocking a replenishment order, is fixed at \$1000 for all products in the primary experiment. The *lost sales cost (B)*, incurred when customer demand cannot be met, is \$102 for Product 1, \$72 for Product 2, \$150 for Product 3, and \$49 for Product 4. The *holding cost (H)*, calculated based on the inventory remaining at the end of each replenishment cycle (overnight in real-world settings), is set at \$0.48 per item per day for Product 1, \$0.40 for Product 2, \$0.54 for Product 3, and \$0.25 for Product 4.

The *lead time*, defined as the duration between placing a replenishment order and its arrival in inventory, is set to 30 days for all experiments. The

replenishment interval refers to the fixed schedule for ordering stock, while the *planning window*, which represents the period over which inventory decisions are optimized, is fixed at 60 days. These parameters are consistent across all trials.

To ensure foresight in decision-making, the planning window incorporates future days even as the trial approaches its end. For instance, on day 110 with a 60-day planning window, decisions account for inventory requirements through day 170, even though the trial concludes on day 180. This approach ensures long-term inventory needs are considered throughout the simulation.

4.1 Design of the simulation trials

Our experiments compare the Static Periodic Review and Forecast-driven Periodic Review replenishment policies against the Predict and Optimize model using scenarios generated by LightGBM forecasts.

The LightGBM models are trained using data from the time period directly preceding the testing interval. This involves fitting the LightGBM model and estimating the parameters needed to generate scenario forecasts. The training setup is consistent across all products. The training period is defined as the 730 days (approximately 2 years) immediately before the test interval. A total of 10 future scenarios are generated for the next W days. The forecast-driven periodic review method compute the man μ_D and standard deviation σ_D from these 10 scenarios each time it is invoked.

For each product, sufficient training and testing data is available to ensure that the test period can begin on any day of the year. This setup avoids bias by ensuring that tests are not concentrated in specific times of the year.

In each trial, the initial orders $\bar{O}$ for the previous l days are all set to 0.

The starting inventory IS is initialized as $IS = \frac{S+s}{2}$, representing an estimate of the average inventory on hand under the assumption that no stock is initially on order.

The simulation runs for 180 days (6 months), aligning with the case study's objective of planning inventory for a 6-month projection period. This information is communicated to suppliers to enable them to adjust their production and inventory levels accordingly. During each replenishment cycle, the original order quantities are maintained, and suppliers reliably deliver the exact amounts ordered. This approach involves creating a 6-month projection using a stochastic model. Inventory decisions during each replenishment cycle are based on the original inventory position and the forecasted future demand, ensuring a dynamic and informed approach to inventory management.

4.2 Outputs

Inventory operating costs (in dollars) are calculated as:

- *Total cost* represents the total of replenishment, holding, and lost sales costs, as defined below.

- *Replenishment cost* Calculates by multiplying the total number of orders by the fixed cost associated with each order. Note that the optimisation model does not necessarily need to order every r days.
- *Holding cost* determines as the product of the inventory remaining at the end of each day and the daily cost per unit. Consequently, the average holding cost aligns with the average inventory level in the summarized results multiplied to holding cost, H, per day per item presented later.
- *Lost sales cost* is the number of lost sales multiplied by the fixed cost incurred for each lost sale.

The other inventory performance and service indicators reported in the experiment sections are calculated as:

- *Lost sales quantity* represents the total unsupplied demand, where we assume customers do not return to make a purchase once the product is restocked.
- *Type 1 service level* is the probability that no lost sales occur during any replenishment cycle, regardless of the quantity.
- *Type 2 service level* refers to the probability of meeting customer demand, which corresponds to the fraction of demand fulfilled without delay during the simulation period [21].
- *Order frequency* is the average number of orders placed during the 120-day evaluation period.
- *Average inventory level* indicates the mean quantity of units stored in the warehouse over the 120-day simulation period.

Computing platform The stochastic optimization model was implemented using the Gurobi 9.0 Mixed Integer Programming solver, with Python 3.9.8 as the interface to manage inputs and iterations. All experiments were conducted on a MacBook Pro equipped with an Apple M1 Chip (8 cores, 3.2 GHz clock speed) and 16 GB RAM.

5 Experimental results

5.1 Main experiment

The results in Table 1 show the performance of the Periodic Review, Forecast-driven Periodic Review and Predict and Optimize approaches for each of the products studied. These results highlight key differences in the performance of each approach, particularly in terms of total cost, order quantities, lost sales quantities, and service levels.

The error of the LightGBM prediction was calculated as the Mean Absolute Percent Error (MAPE) over each 60 day rolling horizon, averaged over the planning period of 180 days. The MAPE for Product 1 is %10.43, Product 2 %14.25, Product 3 %16.01, and Product 4 %22.79. These results show that LightGBM is reasonably accurate for a difficult prediction problem.

Table 1 Results for the periodic review, forecast-driven periodic review, and predict-and-optimize approaches under the cost categories: reorder cost (A), holding cost (H), and lost sales cost (B)

	Product	1	2	3	4
Setup	A	1000	1000	1000	1000
	H	0.48	0.4	0.54	0.25
	B	102	72	150	49
Static periodic review	Total Cost	83,680.33	150,554.92	160,270.37	310,170.53
	Ordering Cost	4000.00	4000.00	4000.00	4000.00
	Holding Cost	12,662.45	25,406.28	19,512.37	29,423.43
	Lost sale Cost	67,017.88	121,148.64	136,758.00	276,747.10
	SL 1	0.74	0.72	0.74	0.63
	SL 2	0.70	0.71	0.74	0.63
	Number of order	4.00	4.00	4.00	4.00
	Lost sales units	657.04	1,682.62	911.72	5,647.90
Forecast-driven periodic review	Total Cost	67,553.72	131,806.95	155,464.41	303,400.10
	Ordering Cost	4000.00	4000.00	4000.00	4000.00
	Holding Cost	13,626.76	22,687.67	20,026.91	31,270.14
	Lost sale Cost	49,926.96	105,119.28	131,437.50	268,129.96
	SL 1	0.79	0.74	0.74	0.64
	SL 2	0.78	0.73	0.74	0.64
	Number of order	4.00	4.00	4.00	4.00
	Lost sales units	489.48	1,459.99	876.25	5,472.04
Predict and optimize	Total Cost	63,840.74	52,664.36	80,476.72	14,4700.61
	Ordering Cost	4000.00	4000.00	4000.00	4000.00
	Holding Cost	14,318.14	30,467.08	25,116.72	43,348.88
	Lost sale Cost	45,522.60	18,197.28	51,360.00	97,351.73
	SL 1	0.81	0.95	0.90	0.87
	SL 2	0.80	0.95	0.90	0.87
	Number of order	4.00	4.00	4.00	4.00
	Lost sales units	446.30	252.74	342.40	1,986.77

It can be seen that total costs for the Predict and Optimize model are consistently lower across all products, compared to the static and forecast-driven periodic review approaches. This cost advantage is primarily driven by the ability of the predict and optimize model to more effectively adapt to the changes in demand as it makes the decisions based on the predicted future demand. This is evident in the predict and optimize model have higher service levels (both Type 1 and Type 2) than both the periodic review and forecast-driven period review. Higher service levels necessitate a moderate increase in holding costs, and result in a greater reduction in lost sales cost across all products.

The statistical significance of improvements in total cost achieved by the *Static Periodic Review* and *Forecast-driven Periodic Review* policies compared to the *Predict and Optimize* model is evaluated using a paired t-test, summarized in Table 2.

The results show a clear statistically significant improvement of the Predict and Optimise model, even in the closest case.

6 Sensitivity analysis

Table 3 presents a sensitivity analysis of inventory management policies under varying cost parameters and demand scenarios. Three policies are evaluated: Periodic review, Predict and Optimize, and Forecast-driven Periodic review. Key metrics include total cost, ordering cost, holding cost, lost sales cost, service levels (SL 1 and SL 2), number of orders, and lost sales quantity. It shows that the Predict and Optimize policy consistently achieves the lowest total costs, ranging from \$50,567.32 to \$72,164.03, demonstrating its cost efficiency compared to the periodic review and forecast-driven periodic review policies, which illustrate higher total costs due to increase lost sales and holding costs. The periodic review policy imposes the highest total costs, peaking at \$95,153.45, primarily driven by significant lost sales costs. The Predict and Optimize policy also effectively mitigates lost sales costs, with values as low as \$31,270.91, and achieves the highest service levels (SL 1 and SL 2 consistently above 0.80), reflecting its ability to meet customer demand effectively. However, the Periodic review and Forecast-driven Periodic review policies show lower service levels (ranging between 0.71 and 0.83) and higher lost sales quantities, with the Periodic review policy experiencing lost sales as high as approximately \$699.12.

The analysis highlights the Predict and Optimize policy as the most reliable and cost-effective approach for managing inventory under stochastic demand conditions. While all policies maintain a fixed number of orders (4.00), the Predict and Optimize policy optimizes order quantities to minimize costs and lost sales, making it particularly effective in environments with high demand variability. The Periodic review policy, with its high lost sales costs and lower service levels, proves less effective, while the Forecast-driven Periodic Review policy offers a middle ground but still falls short of the Predict and Optimize policy's performance. This emphasis the importance of selecting an inventory policy that aligns with demand patterns and cost structures to achieve optimal performance. The Predict and Optimize policy emerges as the preferred strategy for balancing holding and lost sales costs, ensuring robust inventory management under uncertainty.

Table 2 t-Test comparing total costs for static periodic review and forecast-driven periodic review against the predict and optimize approach

	Static periodic review *vs* PO		Forecast-driven periodic review *vs* PO	
	t	*p* value	t	*p* value
Product 1	13.50	3.41×10^{-24}	3.15	2.1×10^{-3}
Product 2	13.64	1.77×10^{-24}	18.16	2.80×10^{-33}
Product 3	43.44	2.78×10^{-66}	22.05	5.42×10^{-40}
Product 4	62.27	3.27×10^{-81}	44.50	2.91×10^{-67}

Table 3 Sensitivity analysis of predict-and-optimize and periodic review approaches for product 1 across reorder cost (A), holding cost(H), and lost sales cost (B)

	Product	1	1	1	1	1	1	1
Setup	A	1000	750	1250	1000	1000	1000	1000
	H	0.48	0.48	0.48	0.36	0.6	0.48	0.48
	B	102	102	102	102	102	76.5	127.5
Static periodic review	Total Cost	83,680.33	86,079.32	82,169.52	70,140.61	89,315.16	63,216.75	95,153.45
	Ordering Cost	4,000.00	3,000.00	5,000.00	4,000.00	4,000.00	4,000.00	4,000.00
	Holding Cost	12,662.45	11,768.88	13,183.49	10,547.35	14,201.78	12,974.33	12,828.16
	Lost sales Cost	67,017.88	71,310.44	63,986.03	55,593.26	71,113.38	46,242.41	78,325.29
	SL 1	0.74	0.72	0.75	0.78	0.71	0.75	0.75
	SL 2	0.70	0.68	0.71	0.75	0.68	0.73	0.72
	Number of order	4.00	4.00	4.00	4.00	4.00	4.00	4.00
	Lost sales units	657.04	699.12	627.31	545.03	697.19	604.48	614.32
Forecast-driven periodic review	Total Cost	67,553.72	68,735.06	65,688.49	55,534.94	71,591.36	54,252.36	75,385.85
	Ordering Cost	4,000.00	3,000.00	5,000.00	4,000.00	4,000.00	4,000.00	4,000.00
	Holding Cost	13,626.76	13,266.26	14,213.21	11,281.66	16,282.30	13,929.40	14,068.23
	Lost sales Cost	49,926.96	52,468.80	46,475.28	40,253.28	51,309.06	36,322.97	57,317.63
	SL 1	0.79	0.78	0.80	0.83	0.79	0.80	0.81
	SL 2	0.78	0.77	0.80	0.82	0.77	0.79	0.80
	Number of order	4.00	4.00	4.00	4.00	4.00	4.00	4.00
	Lost sales units	489.48	514.40	455.64	394.64	503.03	474.81	449.55

Table 3 (continued)

	Product	1	1	1	1	1	1	1
Predict and optimize	Total Cost	63,840.74	63,726.29	65,870.63	55,644.79	67,464.02	50,567.32	72,164.03
	Ordering Cost	4,000.00	3,000.00	5,000.00	4,000.00	4,000.00	4,000.00	4,000.00
	Holding Cost	14,318.14	14,537.63	14,349.45	11,302.77	17,807.80	15,296.41	14,992.71
	Lost sales Cost	45,522.60	46,188.66	46,521.18	40,342.02	45,656.22	31,270.91	53,171.33
	SL 1	0.81	0.81	0.81	0.83	0.81	0.83	0.82
	SL 2	0.80	0.80	0.80	0.83	0.80	0.82	0.82
	Number of order	4.00	4.00	4.00	4.00	4.00	4.00	4.00
	Lost sales units	446.30	452.83	456.09	395.51	447.61	408.77	417.03

The trends observed for all four products are remarkably similar, indicating consistent behavior across the different product categories. To avoid redundancy and maintain conciseness, the sensitivity analysis tables for Product 2, Product 3, and Product 4, detailing their total costs, are provided in Table 4.

6.1 Computation time

The construction of LightGBM models is highly efficient, requiring approximately 583 ms per day to produce a demand forecast for a single product. Generating 10 scenarios and computing the mean and standard deviation for the forecast-driven periodic review model is nearly instantaneous, taking around 28 ms. Solving the stochastic optimization problem is similarly quick takes approximately 1,301 ms using 10 future scenarios for planning window of 60 days. We conducted stochastic optimization experiments with planning windows of 120 and 180 days, which required 3,193 ms and 4,721 ms, respectively. The longer planning window significantly increased computation time, yet the results showed minimal differences.

Table 4 The sensitivity analysis of products 2, 3 and 4 on total cost across reorder cost (A), holding cost(H), and lost sales cost (B)

Setup				Static periodic review	Forecast-driven periodic review	PO
Product	A	H	B	Total Cost	Total Cost	Total Cost
2	1000	0.4	72	150554.92	131806.95	52664.36
2	750	0.4	72	133482.21	123866.92	47303.60
2	1250	0.4	72	133240.07	118354.59	51628.14
2	1000	0.3	72	141917.30	122196.18	46134.79
2	1000	0.5	72	158379.94	139408.42	58920.74
2	1000	0.4	54	109734.29	98643.68	46217.74
2	1000	0.4	90	155393.23	144019.57	51197.76
3	1000	0.54	150	160270.37	155464.41	80476.72
3	750	0.54	150	166135.19	168713.37	79675.52
3	1250	0.54	150	148514.71	150575.48	81767.77
3	1000	0.405	150	137983.64	135418.00	70590.28
3	1000	0.675	150	174642.30	173652.13	89491.93
3	1000	0.54	112.5	126070.30	127723.65	70016.14
3	1000	0.54	187.5	194219.02	185718.77	93527.65
4	1000	0.25	49	310170.53	303400.10	144700.61
4	750	0.25	49	319467.76	315276.57	145252.50
4	1250	0.25	49	295816.71	282104.37	136305.54
4	1000	0.187	49	287497.16	284366.46	132638.51
4	1000	0.3125	49	325945.05	318751.97	153028.11
4	1000	0.25	36.75	238529.51	228408.42	116254.26
4	1000	0.25	61.25	383696.15	370512.16	169161.13

We also experimented with using 100 scenarios for the optimization model to enhance robustness. However, despite the increased scenario count, the performance gains were marginal, with only a slight reduction in total cost for the Predict and Optimize model. The computation times for 100 scenarios takes 8,291 ms. These results indicate that, while using 100 scenarios provides a more robust solution, the additional computational cost may not be justified in this context, as the improvement over using 10 scenarios is minimal.

7 Managerial insights

This study aims to propose an optimal inventory model customized to the case study company, providing more accurate demand predictions and improving their inventory management strategy. The key strength of this approach is its ability to enhance customer service levels efficiently by providing more precise demand forecasting compared to the company's existing model.

While the proposed model offers practical value, several considerations are important for implementation. First, the model assumes access to extensive historical sales data, at least two years, which may not be available for all products in practice. It also operates under a fixed review schedule without accommodating emergency orders, and it assumes a constant ordering cost regardless of order size, which may not reflect real-world cost structures. From a scalability perspective, although the model's computational time increases with the planning horizon, our analysis shows that a 60-day look-ahead window is sufficient for typical reorder intervals (e.g., 30 days), maintaining efficiency. Furthermore, the model scales linearly with the number of products, making it suitable for large product portfolios with the potential for parallel processing. It is worth mentioning that the forecasting component is highly adaptive and data-driven, allowing it to respond effectively to shifts in demand patterns and correct itself after encountering biased or unusual data trends.

Based on the companie's feedback, our model has demonstrated significant advantages as it enables the company to predict future demand more accurately and determine order quantities that reduce lost sales. In some instances, the model offers a broader view of demand trends over a six-month period, supporting better decision-making in order planning. This extended forecasting capability helps optimize inventory levels by striking a balance between the prevention of overstocking and minimization of lost sales.

Furthermore, improved demand visibility strengthens the company's coordination with its suppliers. With a clearer projection of future inventory requirements, the company can ccommunicate its expected demand in advance, allowing suppliers well-prepared to fulfill orders on time. This proactive strategy enhances operational efficiency, fosters supply chain resilience, and ensures timely order fulfillment. Ultimately, our model provides a comprehensive solution that integrates cost-effective inventory management, demand uncertainty mitigation, and service level optimization, making it a valuable tool for long-term business success.

8 Conclusion

In this paper, we demonstrate that inventory management can be significantly improved by moving beyond traditional policy-based approaches and instead using more accurate predictions of future demand. By applying stochastic optimization to make inventory planning decisions based on these forecasts, we achieve lower-cost solutions compared to both static and forecast-driven periodic review models. The key advantage lies in having a clearer picture of future demand and greater flexibility in decision-making.

Our experiments also highlight that effective replenishment decisions can be made quickly, without requiring excessive computation time. The speed at which LightGBM models can be trained allows us to update demand predictions frequently, ensuring they remain responsive to changes in real-time demand patterns. This raises an interesting question: could similar improvements be applied to inventory management systems where predictions are used to set constraints rather than objectives? If so, this might reduce the number of scenarios needed to achieve reliable results, opening the door to solving more complex problems efficiently.

There are two potential extensions of this research. One possibility is extending the current model to order multiple products, enabling coordinated replenishment decisions for groups of items from the same supplier. Another potential extension of this work is to examine its performance in other applications like transportation and routing problems, where demand forecasting and optimization play a critical role.

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Declarations

Declaration of Interest statement The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process During the preparation of this work, the authors did not use any generative AI and AI-assisted technologies in order to prepare the Manuscript.

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